Interpretability Analysis of Feedback Alignment on CNN

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Background

Backpropagation is a fundamental machine learning algorithm for training neural networks but has received scrutiny from computational neuroscientists for its biological implausibility (Grossberg, 1987). Particularly, the Weight Transport Problem raises doubt for any intermediary neuron to acknowledge the weights of those connected from it. Alternative learning algorithms have been proposed to address such limitations and have demonstrated comparable performance with backpropagation on a variety of tasks. A particular class of such algorithms derive from the Feedback Alignment Lillicrap, Cownden, Tweed, and Akerman (2016) algorithm, which uses a fixed matrix B in place of the transposed weight matrix W^T in the backward pass. Interestingly, feedback alignment networks are well known to particularly struggle with convolutional network architectures (Bartunov et al., 2018).

Problem

Despite an abundance of research applying feedback alignment algorithms to various problems and tasks, interpretability analyses of networks trained with feedback alignment appear lacking. We are interested in investigating and elucidating potential differences in feature capture by backpropagation and feedback alignment networks. In particular, we seek to investigate potential structural differences in feedback alignment-trained convolutional networks (CNN) from those trained with backpropagation, a notable limitation for feedback alignment. Aside from potential insight for improvement, studies from neuroscience have shown alignment of deep CNN and the visual cortex across time, (Yamins et al., 2014; Cichy, Khosla, Pantazis, Torralba, & Oliva, 2016). Whether this biological plausible learning algorithm can preserve an architecture's biological plausibility remains unanswered.

Methods

Learning Algorithms

We compare five learning rules that differ only in how the error signal is propagated backward through the network. In all cases, the *weight update* at each layer is computed from the local error signal and the pre-synaptic activations in the standard way; the methods differ solely in how that error signal is obtained.

Backpropagation (BP). Standard backpropagation computes the error signal $\delta^l = \frac{\partial \mathcal{L}}{\partial \mathbf{u}^l}$ at layer l by transposing the forward weight matrix:

$$\delta^l = \left({W^{l+1}}^\top \delta^{l+1} \right) \odot f'\left(\mathbf{u}^l\right), \tag{1}$$

where $\mathcal{L}$ is the loss, $\mathbf{u}^l$ is the pre-activation at layer l, f' is the derivative of the activation function, and $\odot$ denotes the Hadamard product. For convolutional layers, the transpose operation corresponds to a convolution with the 180°-rotated forward kernel $\bar{W}^{l+1}$:

$$\delta_i^l = \left(\bar{W}_i^{l+1} * \delta_i^{l+1} \right) \odot f'\left(\mathbf{u}_i^l\right), \tag{2}$$

where $*$ denotes convolution and i indexes channels. This requires the backward path to have exact knowledge of the forward weights—the "weight transport problem" (Grossberg, 1987; Lillicrap et al., 2016)—which is considered biologically implausible.

Feedback Alignment (FA). Feedback Alignment (Lillicrap et al., 2016) replaces the transposed forward weights with a fixed, random feedback matrix B that is initialized at the beginning of training:

$$\delta^l = \left(B^{l+1} \delta^{l+1} \right) \odot f'\left(\mathbf{u}^l\right). \tag{3}$$

The weight update rule remains identical to BP: $\Delta W^l \propto \delta^l \, {\mathbf{x}^{l-1}}^\top$ for fully-connected layers, and $\Delta W^l \propto \delta^l * \bar{\mathbf{I}}^{l-1}$ for convolutional layers (where $\bar{\mathbf{I}}^{l-1}$ denotes the flipped input activations).

FA in convolutional layers: Dense vs. Toeplitz feedback. The original FA formulation (Lillicrap et al., 2016) was developed for fully-connected networks. When extending FA to convolutional architectures, the form of the feedback operation is not uniquely determined. We identify two natural choices and implement both:

FA (Random) *Dense matrix multiplication.* The feedback is applied as a dense matrix–vector product between a fixed random matrix $B \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and the incoming error signal. This treats the convolutional layer's backward pass as if it were fully-connected, discarding the spatial structure of the convolution entirely.

FA (Toeplitz) *Convolutional feedback.* The feedback is applied as a transposed convolution with a fixed random kernel $B_0 \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k_H \times k_W}$:

$$\tilde{\delta}^l = B_0 *^\top \delta^{l+1}, \tag{4}$$

where $*^\top$ denotes the transposed convolution (with the same padding and stride as the forward layer). We call this the *Toeplitz* variant because a discrete convolution, when expressed as a matrix–vector product, corresponds to multiplication by a doubly block-circulant (Toeplitz-structured) matrix (Gray, 2006). That is, the random feedback kernel B_0 implicitly defines a structured random matrix whose non-zero entries are shared according to the convolutional weight-tying pattern. In backpropagation, this transposed convolution with the *forward* kernel is precisely how gradients flow through convolutional layers; the Toeplitz variant preserves this structural prior while randomizing the kernel entries.

This distinction has practical consequences. The dense feedback matrix has $d_{\text{out}} \times d_{\text{in}}$ free parameters, while the Toeplitz kernel has only $C_{\text{out}} \times C_{\text{in}} \times k_H \times k_W$—substantially fewer for typical convolutional layers, making the Toeplitz variant more computationally efficient.

Unsigned Sign-Flip (uSF) variants. Refinetti, d'Ascoli, Ohana, and Goldt (2021) identified angular alignment between the FA error signal and the true BP gradient as a common feature of successful FA learning, showing that networks first align their weights with the feedback before memorizing the data. Building on this, Moskovitz, Litwin-Kumar, and Abbott (2018) proposed enforcing *sign concordance* between the feedback and forward weights—a lightweight form of alignment—to close the performance gap in deep convolutional networks:

uSF Init: The feedback matrix at training step t is

$$B_t^l = \left|B_0^l\right| \odot \operatorname{sign}\left(W_t^l\right), \tag{5}$$

where B_0^l is the initial random feedback matrix. This retains the random magnitudes from initialization but copies the sign pattern of the current forward weights at each step.

uSF SN (Spectral Normalization): The feedback is

$$B_t^l = \|W_t^l\|_2 \frac{\operatorname{sign}\left(W_t^l\right)}{\|\operatorname{sign}\left(W_t^l\right)\|_2}, \tag{6}$$

which normalizes the feedback to match the spectral norm of the forward weights while using only sign information. Notably, B_t^l under uSF SN is entirely determined by W_t^l—the initial random matrix B_0^l plays no role.

Both uSF variants use the Toeplitz (convolutional) feedback structure for convolutional layers.

Implementation details. The feedback matrices B_0 are drawn from $\mathcal{N}(0, \sigma^2 I)$ where σ is a scaling factor, and remain constant throughout training for the pure FA variants. For the uSF variants, the feedback is recomputed from B_0 and W_t at each backward pass. The FA feedback is applied to all layers except the first convolutional layer, which is the input layer and receives no feedback signal from deeper layers. We adopt Architecture 1 from Moskovitz et al. (2018) which consists of two convolutional layers followed by three Linear ones, with ReLU activations and cross-entropy loss. All methods are trained on CIFAR-10 (Krizhevsky, 2009) with the Adam optimizer (Kingma & Ba, 2015) for 50 epochs and tested on the one with the best validation accuracy. Hyperparameters (learning rate, weight decay, feedback scale) are selected via grid search over 3 epochs, with the best configuration chosen by validation accuracy. All models share the same random seed for comparability. Full architecture, dataset, and hyperparameter details are provided in the Supplementary Information.

Gradient Alignment and Sign Concordance

Gradient alignment. A feature of successful learning with FA is that the forward weights gradually align with the fixed feedback matrix during learning (Lillicrap et al., 2016). We track this alignment online throughout training by computing, at each backward pass and for each layer, the cosine angle (in degrees) between the true BP gradient $\frac{\partial \mathcal{L}}{\partial \mathbf{u}^l}$ and the FA feedback signal $\tilde{\delta}^l$, averaged over the mini-batch. An angle of 90° indicates orthogonality (chance level for high-dimensional random vectors), while decreasing angles indicate alignment.

Sign concordance. Moskovitz et al. (2018) identified sign concordance—the fraction of weight entries where $\operatorname{sign}(W_t^l) = \operatorname{sign}(B_0^l)$—as a key predictor of FA performance, and proposed the uSF variants to enforce it. Since the uSF methods enforce sign agreement by construction, we track this metric only for the unmodified FA variants (FA Random and FA Toeplitz) to investigate whether sign concordance emerges naturally alongside angular alignment, and whether the unmodified methods needed explicit sign enforcement in the first place. For two independent random matrices, the expected sign concordance is 0.5; values significantly above 0.5 indicate that the forward weights have adapted to agree in sign with the feedback.

Centered Kernel Alignment (CKA)

To compare the internal representations learned by different methods, we use linear Centered Kernel Alignment (CKA) (Kornblith, Norouzi, Lee, & Hinton, 2019), which measures similarity between two sets of neural activations in a way that is invariant to orthogonal transformations and isotropic scaling. Our implementation follows Nguyen, Raghu, and Kornblith (2021). For each FA variant, we compute the layer-to-layer CKA matrix between the BP model and the FA model, yielding a 5×5 heatmap (one entry per pair of layers). We evaluate CKA on three data subsets drawn from the combined validation and test sets: (1) all samples, (2) only samples where both BP and the FA method classify correctly, and (3) only samples where BP succeeds but the FA method fails—the subset most informative about where representations diverge.

Feature Visualization

To understand *what* each method's convolutional filters learn to detect, we perform a targeted feature visualization analysis. Rather than examining all 64 channels in each convolutional layer, we design a pipeline that identifies the channels most relevant to a particular classification decision, then characterizes those channels by the real images that maximally activate them.

Focusing on dogs. Preliminary failure analysis revealed that the "dog" class (class 5 in CIFAR-10) is the category where FA (Random) and FA (Toeplitz) exhibit their largest accuracy deficit relative to BP. We therefore focus our feature visualization on the dog class to understand what these methods fail to learn about dog-discriminative features.

Identifying influential channels via GradCAM-style importance. For each method m and each convolutional layer $\ell \in \{\text{conv1}, \text{conv2}\}$, we compute a per-channel importance score adapted from Grad-CAM (Selvaraju et al., 2020). For each dog image $\mathbf{x}_i$ in the test set, we perform a forward pass, recording the activation $A_c^{(\ell)} \in \mathbb{R}^{H \times W}$ at channel c of layer ℓ. We then backpropagate from the model's own predicted class $\hat{y}_i = \arg\max_k f_k(\mathbf{x}_i)$ (not necessarily the ground-truth label) to obtain the gradient $\frac{\partial f_{\hat{y}_i}}{\partial A_c^{(\ell)}}$. The importance of channel c for image i is:

$$\alpha_{c,i}^{(\ell)} = \left| \overline{g}_{c,i}^{(\ell)} \cdot \overline{A}_{c,i}^{(\ell)} \right|, \tag{7}$$

where $\overline{g}_{c,i}^{(\ell)}$ and $\overline{A}_{c,i}^{(\ell)}$ are the spatial means of the gradient and activation maps, respectively. We average this importance over all 1,000 dog test images and select the top 3 channels per layer for detailed visualization.

Note that for fair comparison, all models are loaded with their learned weights into a standard BP architecture, so that the gradient flow used for importance computation is identical across methods—only the learned weights differ.

Top-activating image exemplars. For each of the top-3 channels, we scan the full test set (10,000 images) and find the 9 images that produce the highest mean spatial activation for that channel. By examining which images maximally excite a channel, we obtain an interpretable description of what visual features the channel has learned to detect.

Optimization-based visualization. We also attempted to characterize channels by optimizing a random noise input to maximize channel activation via gradient ascent, with total variation and L2 regularization to encourage natural-looking images. However, the 24×24 input resolution proved too small for the optimized images to produce semantically interpretable patterns, so we rely on the top-activating exemplar approach as our primary visualization method.

Computational Efficiency

The FA variants introduce different computational overheads relative to BP. FA (Toeplitz) is roughly equivalent to BP in cost, since the backward pass performs a transposed convolution with a fixed kernel of the same shape as the forward kernel. FA (Random), by contrast, is the least efficient: its dense feedback matrix for convolutional layers has $d_{\text{out}} \times d_{\text{in}}$ parameters (compared to $C_{\text{out}} \times C_{\text{in}} \times k_H \times k_W$ for the Toeplitz kernel), making its backward pass substantially more expensive in both computation and memory. The uSF variants incur an additional $O(|W|)$ cost per backward pass to recompute the sign-flipped feedback from the current weights, which is modest relative to the forward and backward pass costs. However, uSF Init additionally requires storing the initial random matrix B_0 throughout training, adding $O(|W|)$ extra memory per layer, whereas uSF SN derives the feedback entirely from W_t and requires no auxiliary storage.

Justification and Feasibility

We believe exploring the underlying structure in these networks is a justifiable endeavour because comparing algorithms of differing biological plausibility may lead to novel insights into how the constraints of the weight transport problem dictate the representations a network learns, in addition to why certain architectures like CNN remain limitations for these algorithms. Our conviction in the justifiability of this endeavour is also grounded in the well-established role of interpretability analysis for understanding neural networks and the known limitation of FA to train convolutional neural networks.

We believe the project scope is feasible given the timeframe, as it primarily involves a set of interpretability analyses, following the training of a few, relatively small neural networks. The analysis suite do not require any specific learning algorithm, saving us the time of modification or rewriting. We have designated larger, more challenging datasets as candidate datasets to extend the scope of our project, time permitting. So far, we already have a working version of the model trained on MNIST.

Results

Training Accuracy

Table 1 reports test accuracy for each method. BP achieves 75.80%, and both uSF variants nearly match it (uSF Init: 74.30%, uSF SN: 75.67%). By contrast, FA (Random) and FA (Toeplitz) fall far behind at 35.77% and 40.07%, respectively. The Toeplitz feedback structure provides a modest advantage over the dense random feedback (+4.3 pp), but neither unmodified FA variant approaches BP performance. Figure 1 shows the validation accuracy trajectories: BP and the uSF variants converge to similar levels within 10–15 epochs, while the pure FA methods plateau early at much lower accuracy.

Method	Best Val (%)	Test (%)
BP	75.92	75.80
FA (Random)	36.14	35.77
FA (Toeplitz)	40.02	40.07
uSF Init	74.78	74.30
uSF SN	75.22	75.67

Table 1: Validation and test accuracy on CIFAR-10 (best epoch out of 50).

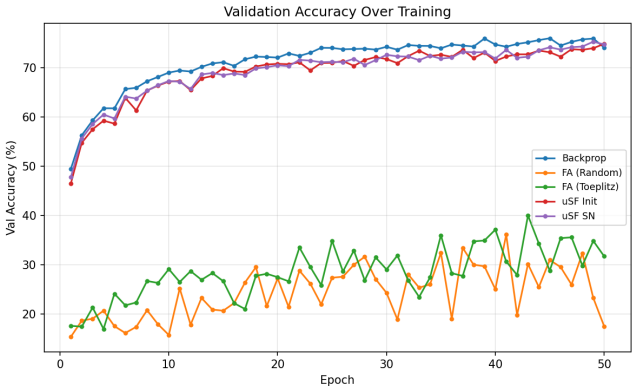

Figure 1: Validation accuracy over 50 epochs of training.

Gradient Alignment

Figure 2 tracks the cosine angle between the true BP gradient and each method's feedback signal during training. All four FA variants start near 90° (chance level.) Over training, the uSF variants rapidly decrease to substantially lower angles, indicating strong alignment with the BP gradient. Consistent with the dynamics described by Refinetti et al. (2021), the alignments then plateau or rises again as the model prioritizes data fitness. The pure FA methods also improve, but their angles remain considerably higher, particularly in the convolutional layer (layer 0). This persistent misalignment in the convolutional layers may partly explain the large accuracy gap between the pure FA variants and the uSF methods.

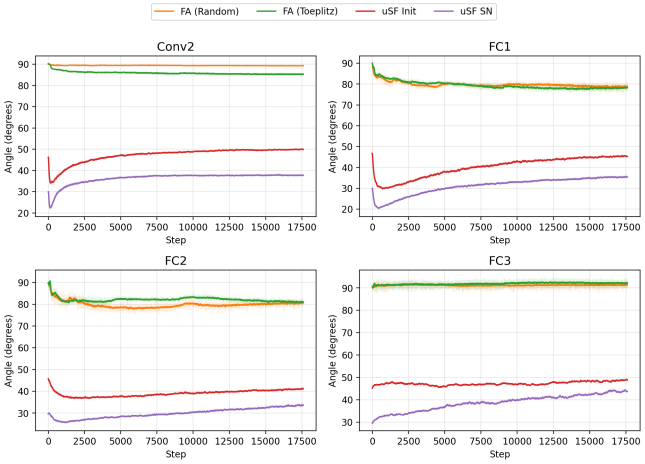

Figure 2: Angle between the BP gradient and the feedback signal during training. Lower angles indicate better alignment. Panels show Conv2, FC1, FC2, and FC3 layers.

Sign Concordance

Figure 3 shows sign concordance between the forward weights and the fixed feedback matrices for the unmodified FA variants at the Conv2 and FC1 layers. Across all layers and both methods, sign concordance remains near the chance level of 0.5 throughout training, with only marginal increases (e.g., FA (Toeplitz) Conv2 rises from 0.500 to 0.514). This indicates that, despite the angular alignment observed in the prior section, the forward weights do not develop meaningful sign agreement with the random feedback matrices. The failure to achieve sign concordance is consistent with the poor performance of the unmodified FA variants and supports the rationale of the uSF methods, which enforce sign agreement by construction. Results for other fully-connected layers (FC2, FC3) are provided in the Supplementary Information and show the same pattern.

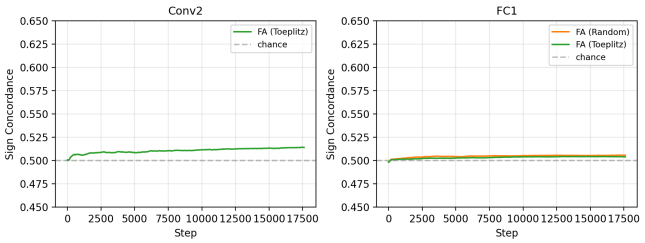

Figure 3: Sign concordance (fraction of elements where $\text{sign}(W) = \text{sign}(B)$) during training for FA (Random) and FA (Toeplitz). The dashed line indicates the 0.5 chance level. Note: FA (Random) does not appear in the Conv2 panel because its dense feedback matrix has a different shape from the convolutional weights, precluding element-wise sign comparison.

Centered Kernel Alignment

Figure 4 shows CKA heatmaps comparing each FA variant's layer-wise representations to BP, evaluated on the subset of images where BP classifies correctly but the FA method fails—the images most informative about representational divergence.

For FA (Random) and FA (Toeplitz), CKA values are uniformly low, even along the diagonal (e.g., conv1–conv1: 0.60 and 0.71, fc3–fc3: 0.07 and 0.15). This confirms that these methods learn fundamentally different representations from BP at every layer, particularly in the deeper fully-connected layers where representations collapse toward near-zero similarity. A striking off-diagonal pattern reinforces this: FA (Random)'s conv2 representations are most similar to BP's *conv1* (CKA = 0.58) rather than BP's conv2 (0.24), and FA (Toeplitz) shows the same trend (0.63 vs. 0.30). This suggests that the pure FA variants fail to learn meaningful additional features in conv2 beyond what conv1 already captures—effectively stalling at the first layer's level of representation.

The uSF variants present a strikingly different picture. Both uSF Init and uSF SN maintain high diagonal CKA values, indicating that even on images they misclassify, these methods learn nearly identical early-layer representations to BP. Similarity decreases in deeper layers but remains substantial, suggesting that classification failures in uSF methods stem from subtle differences in the decision layers rather than fundamentally different feature extraction. CKA results on the full validation dataset and the both-correct subset are provided in the Supplementary Information.

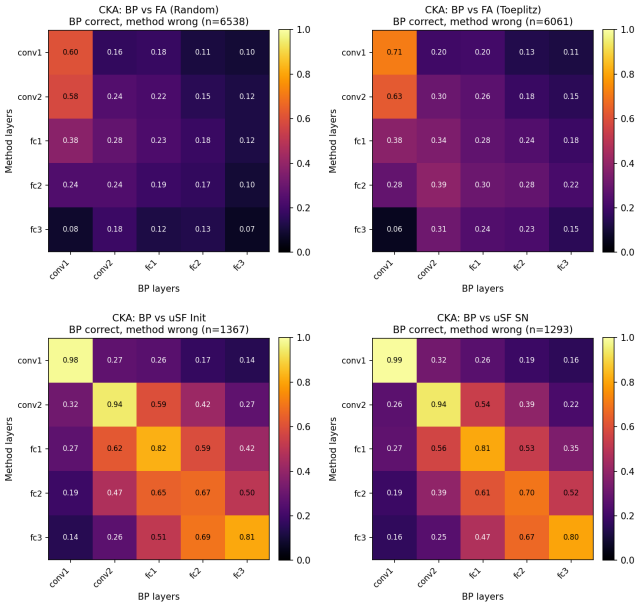

Figure 4: CKA similarity between BP and each FA variant, evaluated on images where BP is correct but the FA method fails. High diagonal values (yellow) indicate layer-matched representations similar to BP.

Feature Visualization

Figure 5 shows the top-activating test images for the three most important Conv2 channels (identified via GradCAM-style importance on dog images) for each method. Each row corresponds to one channel, and each column shows one of the nine images that most strongly activate it.

BP's top Conv2 channels are activated by images of small white dogs (row 1) and dogs with distinctive fur patterns (row 2), demonstrating that BP learns dog-discriminative features at this layer. uSF Init shows a similar pattern: its top channels respond to small dogs and puppies, consistent with the high CKA similarity to BP observed above.

FA (Random) presents a qualitatively different picture. Its top channels are activated by dark, low-contrast images—scenes with dark backgrounds rather than any identifiable object features. This suggests that FA (Random) relies on low-level luminance statistics rather than learning semantic, class-discriminative features in Conv2. FA (Toeplitz) shows slightly more varied activations (responding to bright objects, animals, and vehicles), but without clear dog specificity.

uSF SN's top channels respond to birds and elongated objects (ships, planes), rather than dogs. Despite achieving near-BP accuracy overall, uSF SN appears to route dog classification through different feature pathways than BP.

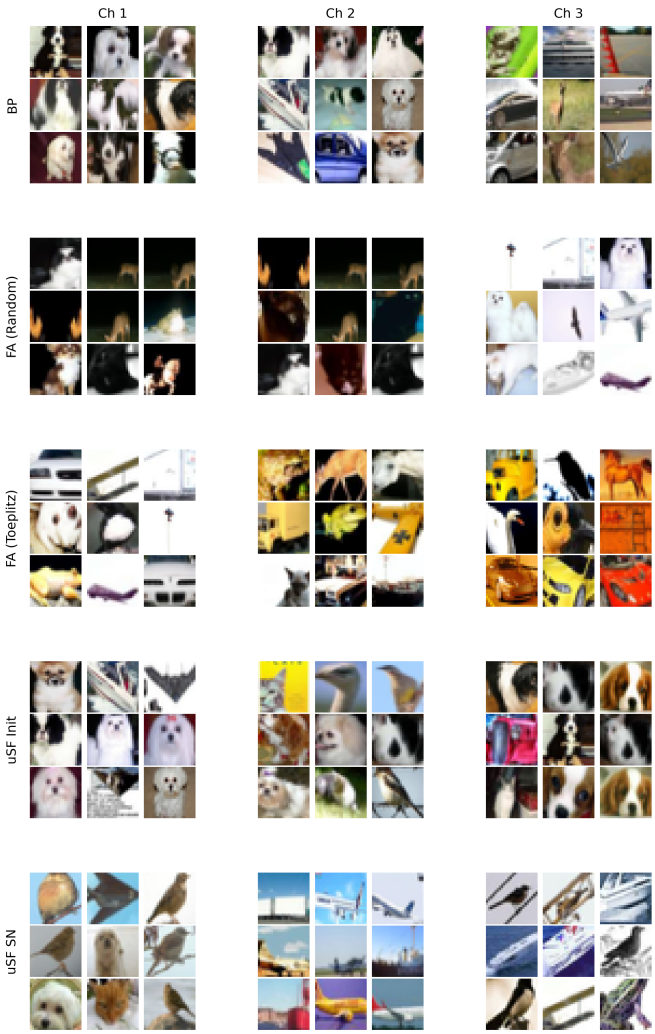

Figure 5: Top-activating test images for the three most dog-important Conv2 channels per method. Each row block is one method (labelled at left); each column block is one channel. Within each block, the 9 highest-activating images are arranged in a 3×3 grid.

Contributions

Larry

1. Responsible for the codebase that includes Torch implementation of FA (Jake's version was in MatLab), training procedure, CKA Analysis and Feature Visualization.

2. Formulate the methodology and pipeline of Feature Visualization.

3. Come up with FA Toeplitz. It's a natural connection, e.g., (Moskovitz et al., 2018) also formulated as such before I read it.

4. Write the Methods and Results section on Training and Interpretability Analysis.

Supplementary Information

Architecture

We adopt Architecture 1 from Moskovitz et al. (2018), a convolutional network designed for $24 \times 24 \times 3$ inputs:

Layer	Configuration
Conv1 + ReLU	$3 \to 64$, 5×5
MaxPool	2×2
Conv2 + ReLU	$64 \to 64$, 5×5
MaxPool	2×2
FC1 + ReLU	$576 \to 384$
FC2 + ReLU	$384 \to 192$
FC3	$192 \to 10$

The model is trained with cross-entropy loss.

Dataset

We use CIFAR-10 (Krizhevsky, 2009), comprising 60,000 32×32 color images in 10 classes. To match the network's 24×24 input size, training images are randomly cropped to 24×24 and randomly flipped horizontally. Validation and test images are center-cropped to 24×24. All images are normalized per-channel (mean $= (0.4914, 0.4822, 0.4465)$, std $= (0.2470, 0.2435, 0.2616)$).

The 50,000 training images are split 90/10 into 45,000 training and 5,000 validation samples (fixed seed $= 42$). The held-out 10,000 test images are used only for final evaluation.

Hyperparameter Search

For each method, we perform a grid search, training for 3 epochs and selecting the configuration with the highest validation accuracy:

- **BP**: learning rate $\in \{5 \times 10^{-4},\ 10^{-3},\ 3 \times 10^{-3}\}$, weight decay $\in \{0,\ 10^{-4}\}$.

- **FA, FA Toeplitz, uSF Init**: learning rate $\in \{10^{-3},\ 3 \times 10^{-3},\ 10^{-2}\}$, feedback scale $\sigma \in \{0.01,\ 0.05,\ 0.1\}$.

- **uSF SN**: learning rate $\in \{10^{-3},\ 3 \times 10^{-3},\ 10^{-2},\ 3 \times 10^{-2}\}$ (feedback scale is irrelevant since B_t^l is fully determined by W_t^l).

All methods are trained with Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$) (Kingma & Ba, 2015) for 50 epochs with batch size 128. The model with the highest validation accuracy is retained for test evaluation. All models share the same initialization seed ($=$ 42).

Sign Concordance (FC Layers)

Figure 6 shows sign concordance for the deeper fully-connected layers (FC2, FC3), complementing the Conv2 and FC1 results in the main text. The same pattern holds: sign concordance remains at or near the 0.5 chance level for both FA (Random) and FA (Toeplitz) throughout training.

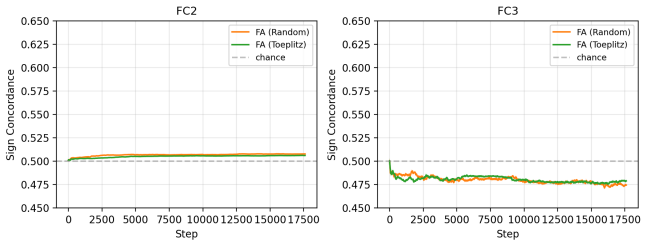

Figure 6: Sign concordance for FC2 and FC3 layers during training.

CKA: Additional Data Subsets

Figures 7 and 8 show CKA heatmaps for the remaining data subsets described in Section : all samples and the both-correct subset, respectively. Both subsets show the same qualitative pattern as the BP-correct subset in the main text: the uSF variants maintain high layer-matched similarity to BP, while the pure FA variants diverge substantially at all layers.

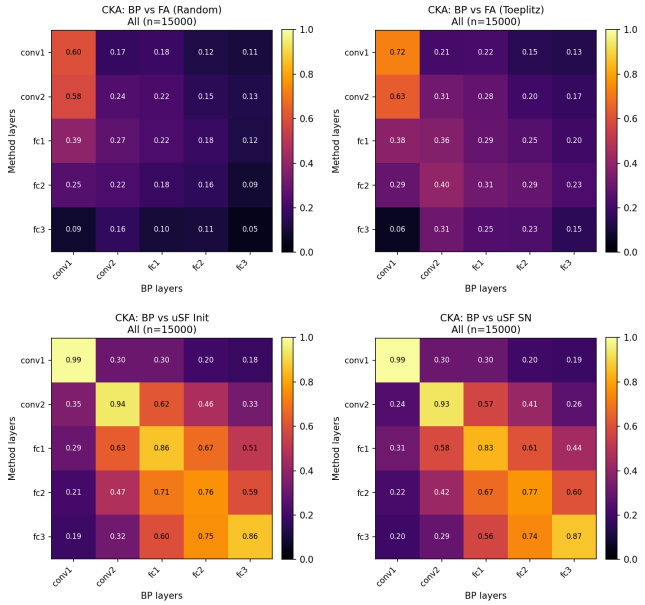

Figure 7: CKA similarity (BP vs. each FA variant) on all validation and test samples.

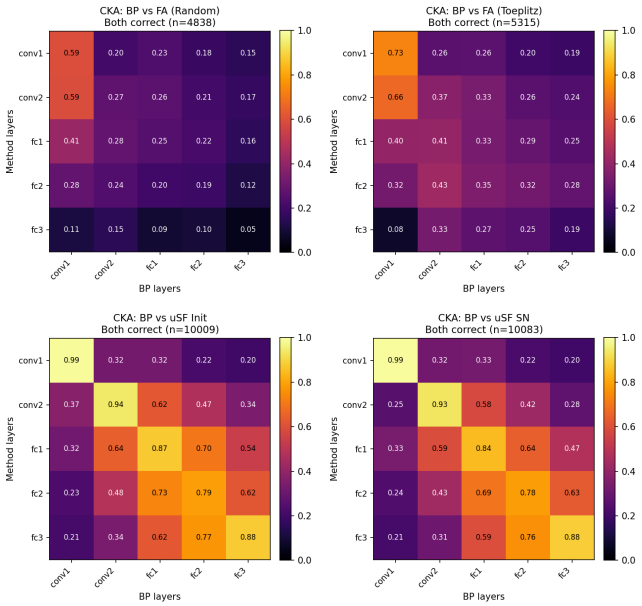

Figure 8: CKA similarity (BP vs. each FA variant) on samples where both methods classify correctly.

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